

Hierarchical Graph-Bag-Network for Self-Supervised Multi-Graph Learning

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Abstract

Multi-Graph Learning (MGL) is a fundamental machine learning paradigm that represents objects as bags-of-graphs, each encoding a distinct structural property, and has broad applications in bioinformatics, chemistry, computing power networks, and software defect detection. However, the inherent scarcity of labeled data poses a significant bottleneck for supervised MGL approaches. While self-supervised contrastive learning offers a compelling solution, its direct application to MGL faces three key challenges: (1) existing graph neural networks, primarily for single-graph modeling, struggle to yield discriminative bag-level representations from bags-of-graphs; (2) conventional contrastive objectives are limited to single-level settings, failing to capture cross-hierarchical dependencies; and (3) standard data augmentation often disrupts intrinsic graph and bag structures, undermining semantic consistency. To address these issues, we propose the Hierarchical Graph-Bag-Network (HGBN), a self-supervised MGL framework that constructs hierarchical representations in the form of a graph-bag-network. HGBN employs an asymmetric hierarchical graph neural network to learn discriminative graph-level and bag-level representations, introduces cross-hierarchical contrastive objectives to align graph-level and bag-level semantics, and leverages the asymmetric network outputs to form positive and negative pairs, preserving intrinsic structural and semantic consistency. Experiments on eight benchmark multi-graph datasets demonstrate that HGBN consistently outperforms both supervised and self-supervised state-of-the-art baselines, achieving average improvements of 4.82% in accuracy and F1 score.

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WWW '26, Dubai, United Arab Emirates.
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ACM ISBN 979-8-4007-2307-0/2026/04
<https://doi.org/10.1145/3774904.3792416>

CCS Concepts

• **Theory of computation** → **Graph algorithms analysis**; • **Information systems** → **Data mining**.

Keywords

Multi-Graph Learning, Self-Supervised Learning, Contrastive Learning, Hierarchical Graph-Bag-Network, Asymmetric Hierarchical Graph Neural Network

ACM Reference Format:

Meixia Wang, Yuhai Zhao, Zhengkui Wang, Fenglong Ma, Yejiang Wang, Miaomiao Huang, Fazal Wahab, Wen Shan, and Xingwei Wang. 2026. Hierarchical Graph-Bag-Network for Self-Supervised Multi-Graph Learning. In *Proceedings of the ACM Web Conference 2026 (WWW '26)*, April 13–17, 2026, Dubai, United Arab Emirates. ACM, New York, NY, USA, 10 pages. <https://doi.org/10.1145/3774904.3792416>

1 Introduction

Graphs serve as an abstraction for modeling complex relational data and have garnered significant attention due to their powerful expressive capacity [6, 17]. Unlike single-instance learning or Multi-Instance Learning (MIL), which model objects as single instances or bags-of-instances, respectively, graph representations are fundamentally suited to modeling the rich structural information inherent in real-world entities. However, single-graph representations are often inadequate for modeling the diverse and intricate structural characteristics of real-world objects [15, 19]. To mitigate this limitation, Multi-Graph Learning (MGL) has been introduced, where objects are modeled as bags-of-graphs with each graph encapsulating complementary structural or semantic information [34, 37]. Compared to single-instance, single-graph [41], or MIL paradigms [27], MGL offers greater flexibility and expressive power, allowing for the more comprehensive modeling of objects with complex structures. It has gained increasing attention and proven effective in diverse domains, including bioinformatics, chemistry, computing power networks, and software defect detection [32, 42].

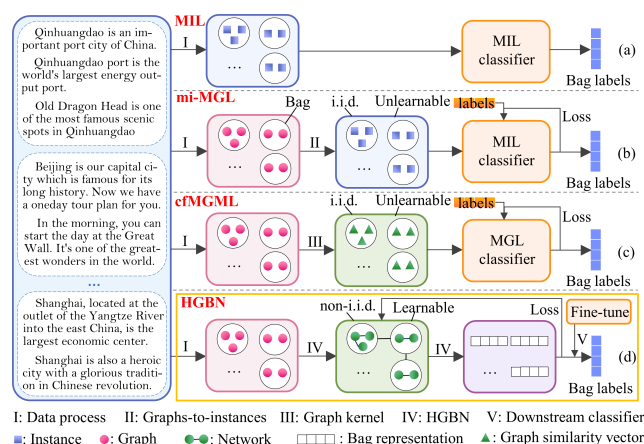


Figure 1: The example of classical MIL and MGL methods for text datasets. (a) Multi-instance learning method (MIL). (b) Multi-instanced MGL method (mi-MGL). (c) Graph kernel-based MGL method (cfMGML). (d) The proposed self-supervised MGL method (HGBN).

Existing MGL methods can be broadly categorized into two classes. The first category, multi-instanced MGL approaches (as illustrated in Figure 1(b)), reduces MGL to an MIL problem and directly adopts conventional MIL techniques (as illustrated in Figure 1(a)) for solution [35, 36, 38]. The second category, graph kernel-based MGL approaches (as illustrated in Figure 1(c)), exploits graph kernels to preserve the intrinsic structural information of individual graphs and address the MGL problem more directly [13, 28, 32, 46]. Although both categories have achieved certain success in MGL tasks, they usually rely heavily on large amounts of labeled data, which is often expensive and impractical to obtain, thereby limiting the flexibility and applicability of MGL in real-world scenarios. To reduce reliance on annotated data, Self-supervised Contrastive Learning (SCL) [16, 23, 33] has emerged as a promising approach. By using pretext tasks as supervisory signals, it learns discriminative representations from unlabeled data and has been widely applied in computer vision, natural language processing, and single-graph learning [3, 12, 29]. However, extending SCL to the MGL setting remains largely unexplored and faces three non-trivial challenges.

The first challenge lies in learning discriminative bag-level representations from bags-of-graphs. This task is particularly challenging because bags-of-graphs naturally exhibit complex hierarchical structures, where graphs within a bag as well as graphs across different bags can share semantic or structural correlations [35, 38]. Capturing these intra-bag and inter-bag dependencies is essential to form meaningful bag-level representations that reflect the overall semantics of the data. Existing MGL approaches, including multi-instanced MGL methods and graph kernel-based MGL methods, attempt to address this challenge but exhibit two critical limitations. One limitation is that they often assume graphs are independently and identically distributed (i.i.d.) both within and across bags, overlooking important inter-graph correlations. Another limitation is that they typically adopt a two-stage paradigm where bag-level features are precomputed and fixed, while only the downstream

classifier is trainable. As a result, the features themselves cannot be jointly optimized for the task. These limitations highlight the necessity of a learnable framework that can intrinsically model hierarchical dependencies from graphs to bags and generate discriminative bag-level representations.

The second challenge is to capture cross-hierarchical dependencies via contrastive objectives. In MGL, a bag’s discriminative semantics are not simply the sum of its constituent graphs; critical information is encoded in the relationships between individual graphs and their enclosing bag. For instance, consider a chemical molecule (bag) and its toxicity. The toxicity may not be determined by any single substructure (graph) in isolation, but by specific combinations of substructures and their interactions with the overall molecule. Conventional contrastive learning objectives [31, 45], which operate solely within a single level (e.g., graph-graph or bag-bag), cannot model such cross-level relationships. While they can distinguish different bags or graphs, they fail to capture how individual graphs contribute to the semantics of their containing bag. Consequently, directly applying standard contrastive learning results in the loss of critical discriminative information. This motivates the development of contrastive objectives that explicitly encode cross-hierarchical dependencies.

The third challenge is to avoid disrupting the intrinsic structural and semantic properties of graphs and bags. Conventional augmentation strategies for single-graph learning, such as random edge removal or feature perturbation, are well-known to perturb the graph topology and undermine its semantic integrity [31]. In the MGL setting, these distortions are exacerbated at the bag level. This is because perturbations in individual graphs are propagated and compounded during the aggregation process, leading to a significant corruption of the bag-level semantics. This issue highlights the critical need for an alternative paradigm that can generate positive and negative pairs in a manner that preserves the structural and semantic integrity of bags and graphs.

To address these challenges, we propose the Hierarchical Graph-Bag-Network (HGBN), a novel self-supervised MGL framework that constructs hierarchical representations in the form of a graph-bag-network. This hierarchical structure integrates three interconnected spaces, namely graph, bag, and network, capturing graph-graph, graph-bag, and bag-bag relationships in a unified representation. Leveraging this structure, HGBN employs an Asymmetric Hierarchical Graph Neural Network (AHGNN) to learn discriminative representations at both graph and bag levels, effectively addressing the challenge of learning bag-level representations from permutation-invariant graph sets. To capture cross-hierarchical dependencies, HGBN introduces a novel cross-hierarchical contrastive objective that aligns graph-level and bag-level semantics. Finally, instead of relying on standard augmentation strategies that may distort intrinsic structures, HGBN forms positive and negative pairs from asymmetric network outputs, preserving both structural and semantic integrity of both graphs and bags. The main contributions of our work are summarized as follows:

- We introduce HGBN, a novel self-supervised learning framework specifically designed for MGL. To the best of our knowledge, this is the first study on self-supervised representation learning for the MGL paradigm.

- We propose a hierarchical graph-bag-network representation that unifies complex multi-graph structures into a single hierarchical framework, and introduce a novel AHGNN method to learn discriminative bag-level representations.
- We design cross-hierarchical contrastive objectives and a novel asymmetric strategy for positive-negative pair generation, which collectively preserve the inherent structural and semantic integrity of multi-graph data.
- Extensive experiments on eight multi-graph datasets demonstrate that HGBN consistently and significantly outperforms state-of-the-art supervised and self-supervised baselines.

2 Related Work

Multi-Graph Learning. MGL was first introduced by Wu et al. [34] as a paradigm where objects are represented as bags-of-graphs, each capturing a distinct structural or semantic property. This allows modeling complex objects with multiple complementary substructures. Early MGL methods (Figure 1(b)), inspired by MIL, treated each graph as an instance, using subgraph mining to construct substructure vocabularies [36, 38] and encode each bag as a binary vector. While simple, these approaches often neglect structural dependencies, leading to weak bag representations. To address this, graph kernel methods [46] (Figure 1(c)) were introduced, mapping graphs into high-dimensional spaces to preserve structural information, achieving success in several applications [13, 28, 32]. However, most MGL methods remain supervised and rely heavily on labeled data, limiting scalability and motivating self-supervised solutions.

Self-Supervised Graph Learning. Graph representation learning excels at modeling complex relational structures [26], but its scalability is hindered by the need for costly annotations [18]. Self-Supervised Learning (SSL) addresses this by leveraging intrinsic graph structures to learn representations without labels [20, 39]. Early SSL methods, such as matrix factorization [2], random walks [11], and graph autoencoding [5], capture structural dependencies but struggle with scalability and generalization. Graph contrastive learning [43, 44] aligns augmented views of the same graph while separating different graphs. Notable variants include SimGRACE [40], which perturbs model parameters, GALOPA [30], which selects consistent positive pairs via optimal transport, and PaGCL [45], which leverages temporal dynamics. Despite their success, existing SSL methods are primarily designed for single-graph settings, and extending them to MGL remains an open challenge.

3 Methodology

To facilitate understanding, we provide an overview of the proposed HGBN framework, as illustrated in Figure 2. HGBN consists of three key components: a hierarchical representation, an AHGNN, and a cross-hierarchical contrastive loss. (1) The hierarchical representation module unifies multi-graph bags into a single computational graph by explicitly constructing a graph-bag-network. This hierarchical representation integrates three interconnected spaces: the graph space, encoding structural information from individual graphs; the bag space, modeling graph-bag interactions; and the network space, capturing higher-order graph-graph, graph-bag, and bag-bag dependencies. (2) To exploit this structure, HGBN employs an AHGNN composed of a graph encoder, a network encoder,

and a momentum encoder. The graph encoder generates graph embeddings, which are further refined through the network and momentum encoders. This asymmetric hierarchical design enables the learning of discriminative representations at both graph and bag levels. (3) To capture cross-hierarchical dependencies, HGBN introduces a cross-hierarchical contrastive objective that aligns graphs with their enclosing bags. Specifically, positive pairs are formed between each graph and its associated bag, while negative pairs are constructed across bags. This strategy explicitly optimizes graph-bag associations while preserving structural and semantic integrity without relying on conventional data augmentation.

3.1 Problem Statement

We study the task of MGL, which aims to map a set of bags, each composed of multiple graphs, to their associated labels. Formally, let $\mathcal{B} = \{B_1, B_2, \dots, B_m\}$ denote the collection of bags. Each bag $B_i = \{g_{i1}, g_{i2}, \dots, g_{in_i}\}$ contains a variable number of graphs, where n_i may differ across bags. Each graph g_{ij} is represented as $g_{ij} = (V_{ij}, A_{ij}, X_{ij})$, where V_{ij} is the node set, $A_{ij} \in \{0, 1\}^{|V_{ij}| \times |V_{ij}|}$ is the adjacency matrix, and $X_{ij} \in \mathbb{R}^{|V_{ij}| \times d_r}$ is the node feature matrix. We consider an SSL setting in which the bag set $\mathcal{B}$ is available without labels. The pretext task is to learn a mapping function $\phi : \mathcal{B} \rightarrow \mathbb{R}^d$ that encodes each bag into a discriminative bag-level representation. In the downstream task, only a subset of bags $\mathcal{Q} \subset \mathcal{B}$ is annotated with labels from $\mathcal{Y}$. The ultimate objective is to leverage the learned representations to train a lightweight classifier on $\mathcal{Q}$ and predict the labels of the remaining unlabeled bags $\bar{\mathcal{Q}} = \mathcal{B} \setminus \mathcal{Q}$.

3.2 Hierarchical Representation

The success of a self-supervised MGL framework critically depends on its ability to learn discriminative bag-level representations that comprehensively capture the rich information encoded in multi-graph data. However, the unique and complex structure of MGL poses significant challenges. Beyond the intrinsic properties of individual graphs and their dependencies with their bag, essential correlations also exist among graphs and across bags. Nevertheless, existing MGL methods often assume graphs and bags are i.i.d. and adopt a two-stage paradigm with fixed bag-level features, limiting their ability to capture dependencies and jointly optimize representations [35, 38]. Furthermore, existing graph representation learning methods, primarily designed for single graphs, are inadequate to handle the complexities of multi-graph data. To address these limitations and natively model the inherent multi-level dependencies, we propose a novel hierarchical graph-bag-network representation. This representation unifies multi-graph bags into a coherent computational structure, explicitly capturing all levels of associations and thus establishing the foundation for learning truly discriminative representations.

Specifically, the proposed hierarchical graph-bag-network representation is composed of three distinct yet interconnected spaces: the graph, bag, and network spaces. The graph space serves as the foundation, encoding structural information at the graph level. It is defined as a collection of discrete graphs, as shown in Eq. (1). Here, g_k is the k -th graph in the graph space $\mathcal{G}$, V_k is its set of nodes, X_k is the node feature matrix, and A_k is the adjacency matrix.

$$\mathcal{G} = \{g_k = (V_k, A_k, X_k) \mid k = 1, 2, \dots, |\mathcal{G}|\} \quad (1)$$

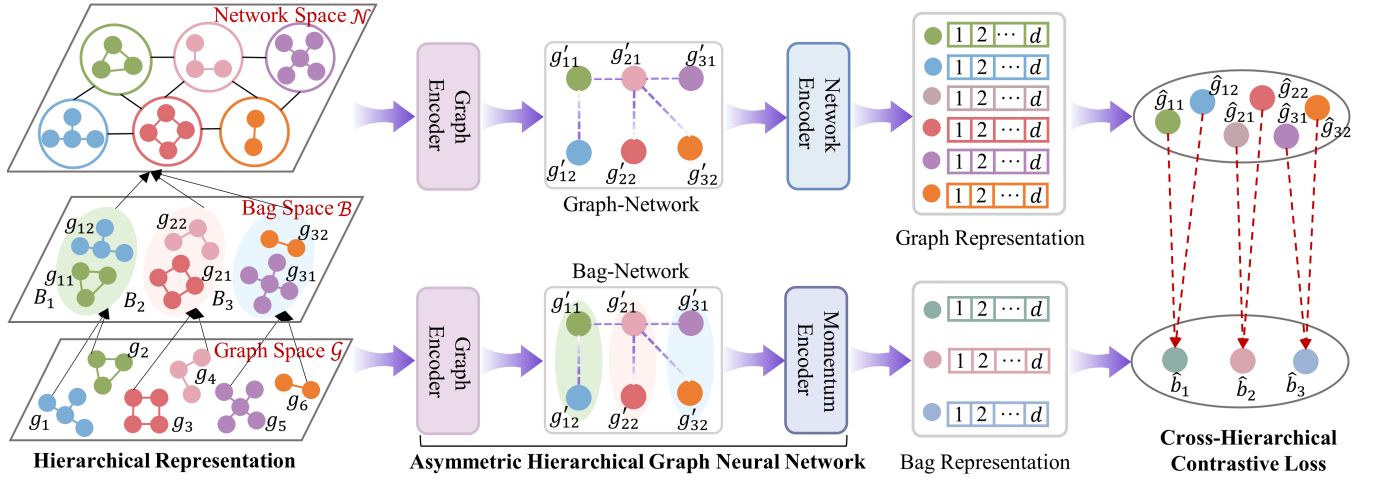


Figure 2: An Overview of HGBN.

The bag space models graph-bag relationships by organizing graphs into their corresponding bags, as shown in Eq. (2).

$$\mathcal{B} = \{B_i | B_i = \{g_{i1}, g_{i2}, \dots, g_{in_i}\}, g_{ij} \in \mathcal{G}, i = 1, 2, \dots, m\} \quad (2)$$

In Eq. (2), $\mathcal{B}$ denotes the bag space containing m bags, where each bag B_i comprises a set of graphs $\{g_{ij}\}_{j=1}^{n_i}$ drawn from the graph space $\mathcal{G}$, with n_i indicating the number of graphs in B_i . By organizing graphs into bags, we establish a hierarchical graph-bag representation that explicitly models both the intrinsic structural information of individual graphs and the dependencies between graphs and their corresponding bag.

Beyond the graph's intrinsic information and the graph-bag relationships, dependencies among graphs and across bags are essential for learning discriminative representations. To explicitly capture these relationships, we introduce a network space built atop the graph and bag spaces. In this space, each graph is treated as a node, and inter-graph as well as inter-bag dependencies are encoded as edges, forming a higher-order relational network. Each node encodes the structural information of its corresponding graph, while each edge models relational dependencies among graphs and their respective bags. The network space is defined in Eq. (3).

$$\mathcal{N} = \{\mathcal{V}, \mathcal{E}\} \quad (3)$$

Here, $\mathcal{V} = \{g_1, g_2, \dots, g_{|\mathcal{V}|}\}$ is the set of graph nodes, and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ denotes edges encoding inter-graph and inter-bag dependencies.

Specifically, each edge in $\mathcal{E}$ directly encodes the dependency between a pair of graphs, g_p and g_q , as defined in Eq. (4). Dependencies between bags are then derived from the edges among their constituent graphs. For instance, the dependency between bag B_i and bag B_j is computed by aggregating all pairwise dependencies between graphs in B_i and graphs in B_j , as formalized in Eq. (5).

$$\mathcal{E}_{\text{graph}}[p, q] = r(g_p, g_q) \quad (4)$$

$$\mathcal{E}_{\text{bag}}[i, j] = \text{Agg}(\{r(g_{ik}, g_{jl}) \mid g_{ik} \in B_i, g_{jl} \in B_j\}) \quad (5)$$

Here, $r(\cdot, \cdot)$ is a relational function between two graphs, and $\text{Agg}(\cdot)$ uses average pooling to aggregate relational information from all graph pairs across two bags into a bag-level dependency score.

To balance learnability and computational efficiency, the edge weights in $\mathcal{N}$ are initially randomly initialized, as shown in Eq. (6), and are subsequently learned and refined during training based on the learned graph representations.

$$\mathcal{E} \sim \text{Uniform}(0, 1) \quad (6)$$

By unifying the graph, bag, and network spaces, our framework establishes a complete hierarchical representation, denoted as $\mathcal{H}$, that provides a comprehensive foundation for self-supervised learning on multi-graph data as shown in Eq. (7).

$$\mathcal{H} = \{\mathcal{G}, \mathcal{B}, \mathcal{N}\} \quad (7)$$

3.3 Asymmetric Hierarchical Graph Neural Network

Conventional GNNs are primarily designed for single-graph data and are thus inadequate for modeling the complex structure of multi-graph bags. Moreover, the hierarchical graph-bag-network explicitly encodes inter-graph and inter-bag dependencies, which cannot be directly processed by standard GNN architectures. To address this, we propose the AHGNN. It employs a graph encoder, a network encoder, and a momentum encoder to jointly learn both graph and bag representations while preserving hierarchical dependencies. This asymmetric hierarchical design enables the extraction of discriminative representations essential for self-supervised MGL.

The AHGNN's first component is the graph encoder $f_g(\cdot)$, which produces informative embeddings that capture the intrinsic node features and topological relationships of individual graphs. Specifically, for a graph $g = (V, \mathbf{A}, \mathbf{X})$, the graph encoder generates a graph embeddings $\mathbf{g}'$ as shown in Eq. (8).

$$\begin{aligned} \mathbf{h}_{v_g}^1 &= \sigma_0(\mathbf{A}' \mathbf{X} \mathbf{W}^0) \\ \mathbf{h}_{v_g}^{l+1} &= \sigma_l(\mathbf{A}' \mathbf{h}_{v_g}^l \mathbf{W}^l), l \in \{1, 2, \dots, L-1\} \\ \mathbf{g}' &= \sum_{v_g \in V} \mathbf{h}_{v_g}^L \end{aligned} \quad (8)$$

In Eq. (8), $\mathbf{A}'$ is the non-negative, symmetric, and normalized adjacency matrix of graph g . $\mathbf{h}_{v_g}^l \in \mathbb{R}^{|V| \times d_l}$ denotes the output of the

$(l - 1)$ -th layer, where d_l is the feature dimension. $\mathbf{W}^l \in \mathbb{R}^{d_l \times d_{l+1}}$ is the learnable weight matrix, and $\sigma_l(\cdot)$ is the nonlinear activation function. The final graph embedding $\mathbf{g}'$ is obtained by aggregating the node-level representations.

To preserve the intrinsic structures of graphs and bags, AHGNN adopts an asymmetric architecture that independently learns discriminative representations at the graph and bag levels. This design enables the construction of positive and negative pairs for cross-hierarchy contrastive learning without relying on conventional data augmentation and mitigates potential model collapse that may arise from improperly defined pairs in symmetric architectures. Based on the graph embeddings $\mathbf{g}'$ from Eq. (8), the hierarchical representation is decomposed into two subnetworks: the graph-network $\mathcal{N}_g$ and the bag-network $\mathcal{N}_b$. These subnetworks allow the asymmetric encoders to separately learn discriminative graph and bag representations while preserving hierarchical dependencies, as shown in Eqs. (9) and (10).

$$\mathcal{N}_g = (\mathcal{G}', \mathcal{E}_{\text{graph}}) \quad (9)$$

$$\mathcal{N}_b = (\mathcal{B}', \mathcal{E}_{\text{bag}}) \quad (10)$$

Here, $\mathcal{G}'$ represents the graph representation space, where each original graph $g_k = (V_k, A_k, X_k) \in \mathcal{G}$ is encoded as $\mathbf{g}'_k$ via the graph encoder, as formalized in Eq. (11).

$$\mathcal{G}' = \{\mathbf{g}'_1, \mathbf{g}'_2, \dots, \mathbf{g}'_{|\mathcal{G}|}\} \quad (11)$$

Similarly, the bag representation space $\mathcal{B}'$ is formed by a collection of bag-of-graph embeddings, where each bag $B_i \in \mathcal{B}$ is encoded by the graph encoder into B'_i , as formalized in Eq. (12).

$$\mathcal{B}' = \{B'_i \mid B'_i = \{\mathbf{g}'_{i1}, \mathbf{g}'_{i2}, \dots, \mathbf{g}'_{in_i}\}, \mathbf{g}'_{ij} \in \mathcal{G}', i = 1, 2, \dots, m\} \quad (12)$$

The edge sets $\mathcal{E}_{\text{graph}}$ and $\mathcal{E}_{\text{bag}}$, encode inter-graph and inter-bag dependencies, respectively. Specifically, the edge weight between two graphs is defined by the similarity of their embeddings, as formalized in Eq. (13). Similarly, the edge weight between two bags is computed based on the similarities among the graphs they contain, as defined in Eq. (14).

$$\mathcal{E}_{\text{graph}}[i, j] = s(\mathbf{g}'_i, \mathbf{g}'_j) \times I[s(\mathbf{g}'_i, \mathbf{g}'_j) \geq \sigma], \mathbf{g}'_i, \mathbf{g}'_j \in \mathcal{G}' \quad (13)$$

$$\mathcal{E}_{\text{bag}}[i, j] = s(B'_i, B'_j) \times I[s(B'_i, B'_j) \geq \sigma], B'_i, B'_j \in \mathcal{B}' \quad (14)$$

Here, $s(\cdot, \cdot)$ is the similarity function defined in Eqs. (15) and (16). $I[\cdot]$ is an indicator function, and $\sigma \in [0, 1]$ is a threshold parameter used to regulate the structural complexity of the constructed network.

$$s(\mathbf{g}'_i, \mathbf{g}'_j) = \mathbb{E}_{\mathbf{w}}[\cos_{\mathbf{w}}(\mathbf{g}'_i, \mathbf{g}'_j)] \quad (15)$$

$$s(B'_i, B'_j) = \text{Agg}(\{s(\mathbf{g}'_{ik}, \mathbf{g}'_{jl}) \mid \mathbf{g}'_{ik} \in B'_i, \mathbf{g}'_{jl} \in B'_j\}) \quad (16)$$

where $\cos_{\mathbf{w}}(\cdot, \cdot)$ denotes weighted cosine similarity with learnable weight $\mathbf{w}$ initialized from a Gaussian distribution.

To derive discriminative graph and bag representations, AHGNN employs a network encoder f_n and a momentum encoder f_m that process the graph-network and bag-network, respectively. As formalized in Eqs. (17) and (18).

$$\begin{aligned} \mathbf{h}_{g'}^1 &= \sigma_0(\mathcal{E}_{\text{graph}} \mathbf{g}' \mathbf{W}_b^0) \\ \mathbf{h}_{g'}^{l+1} &= \sigma_l(\mathcal{E}_{\text{graph}} \mathbf{h}_{g'}^l \mathbf{W}_b^l), l \in \{1, 2, \dots, L_g - 1\} \\ \hat{\mathbf{g}} &= \mathbf{h}_{g'}^{L_g} \end{aligned} \quad (17)$$

$$\begin{aligned} \mathbf{h}_{g''}^1 &= \sigma_0(\mathcal{E}_{\text{bag}} \mathbf{g}' \mathbf{W}_b^0) \\ \mathbf{h}_{g''}^{l+1} &= \sigma_l(\mathcal{E}_{\text{bag}} \mathbf{h}_{g''}^l \mathbf{W}_b^l), l \in \{1, 2, \dots, L_b - 1\} \\ \hat{\mathbf{b}} &= \max_{g'' \in B} \mathbf{h}_{g''}^{L_b} \end{aligned} \quad (18)$$

In Eq. (17), $\mathbf{h}_{g'}^l$ denotes the node embeddings at l -th layer, and $\mathbf{W}_b^l$ is the learnable weight matrix of the l -th layer. The final graph embedding $\hat{\mathbf{g}} \in \mathbb{R}^d$ is obtained from the output of f_n . Similarly, in Eq. (18), $\mathbf{W}_b^l$ is the learnable weight matrix of l -th layer, and $\mathbf{h}_{g''}^l$ denotes the embedding of graph g'' within bag B at l -th layer. The discriminative bag representation $\hat{\mathbf{b}} \in \mathbb{R}^d$ is obtained via max-pooling over the graph within B , following the common MGL assumption that a bag's label is determined by its most representative graph.

Although both encoders share the same architecture, their optimization mechanisms differ: the network encoder is trained via standard backpropagation, whereas the momentum encoder is updated using a momentum-based moving average of the network encoder's parameters. This update rule is defined as Eq. (19).

$$\theta_m = \rho \theta_m + (1 - \rho) \theta_n \quad (19)$$

In Eq. (19) θ_m and θ_n are the parameters of the momentum and network encoders, respectively, and ρ is a hyperparameter, typically set to 0.99. This design stabilizes training, mitigates representation collapse, and improves the consistency among embeddings.

3.4 Cross-Hierarchical Contrastive Loss

Effective self-supervised MGL requires not only a suitable architecture for extracting discriminative bag representations but also a contrastive objective that captures cross-hierarchical dependencies. In MGL, a bag's semantics are not merely the aggregation of its constituent graphs but also emerge from the interactions between individual graphs and their enclosing bag. Conventional contrastive learning, which operates solely within a single level, fails to encode such cross-level relationships and may thus discard critical discriminative information. To address this limitation, we design a cross-hierarchical contrastive loss that explicitly aligns representations across levels. Specifically, we adopt bag representations $\hat{\mathbf{b}}_i$ generated by a momentum encoder as anchors, and graph representations $\hat{\mathbf{g}}_{ij}$ encoded by the network as positive or negative samples relative to these anchors. Graphs within the anchor bag B_i are treated as positives, while graphs from other bags serve as negatives. This construction eliminates the need for conventional data augmentation, thereby avoiding structural distortions of graphs and bags. The loss is defined as Eq. (20).

$$\mathcal{L} = \frac{-1}{m} \sum_{i \in I} \frac{1}{n_i} \sum_{j \in n_i} \log \frac{\exp(\text{sim}(\hat{\mathbf{b}}_i, \hat{\mathbf{g}}_{ij})/\tau)}{\sum_{a \in A(i)} \sum_{j \in n_a} \exp(\text{sim}(\hat{\mathbf{b}}_i, \hat{\mathbf{g}}_{aj})/\tau)} \quad (20)$$

where $I = \{1, 2, \dots, m\}$, $A(i) = I \setminus \{i\}$. The $\text{sim}(\mathbf{x}, \mathbf{y}) = \mathbf{x}^T \mathbf{y} / \|\mathbf{x}\| \|\mathbf{y}\|$ and the τ is a temperature parameter.

3.5 Computational Complexity

The time complexity of HGBN consists of five parts. The graph encoder requires $O(L(N_g D_g^2 + E_g D_g))$, while the network and momentum encoders require $O(L_b(N D_b^2 + E D_b))$, where N and E denote the numbers of nodes and edges in the corresponding graph or bag-level networks, and D_g and D_b represent the feature dimensions.

Table 1: Comparative results on NCI(1), NCI(109), AICV, and AIDB datasets. Best and second-best values in each column are highlighted in bold and underlined. ACC denotes accuracy, and F1 denotes the F1 score.

Algorithms	NCI(1)		NCI(109)		AICV		AIDB	
	ACC (%)	F1 (%)	ACC (%)	F1 (%)	ACC (%)	F1 (%)	ACC (%)	F1 (%)
mi-SVM	54.76 ± 1.65	60.60 ± 1.70	58.18 ± 1.36	57.29 ± 1.03	79.04 ± 0.50	82.20 ± 0.35	77.52 ± 0.39	81.27 ± 0.24
MI-SVM	53.08 ± 2.35	20.99 ± 11.29	54.08 ± 1.40	37.42 ± 7.28	85.21 ± 0.18	84.00 ± 0.18	80.66 ± 0.24	78.73 ± 0.29
MInD	55.97 ± 0.92	56.59 ± 1.09	57.23 ± 1.23	59.40 ± 1.70	90.74 ± 0.34	90.84 ± 0.29	89.38 ± 0.22	89.45 ± 0.30
MILES	55.35 ± 2.18	62.37 ± 3.52	57.98 ± 1.46	63.76 ± 1.61	89.50 ± 0.20	89.67 ± 0.14	88.13 ± 0.03	88.23 ± 0.03
Attention	54.70 ± 0.50	61.20 ± 3.00	54.70 ± 1.40	56.60 ± 2.80	88.80 ± 0.20	88.80 ± 0.20	85.70 ± 0.50	85.90 ± 0.60
G-Attention	55.70 ± 0.90	59.20 ± 2.70	55.30 ± 0.80	56.50 ± 2.50	89.40 ± 0.20	89.50 ± 0.20	85.80 ± 0.40	86.30 ± 0.30
B-Graph	55.68 ± 6.89	56.92 ± 7.73	52.33 ± 7.44	54.86 ± 9.11	90.37 ± 2.13	90.19 ± 1.71	87.27 ± 2.43	87.00 ± 1.41
GraphCL	64.05 ± 6.57	59.57 ± 15.71	61.83 ± 6.35	58.64 ± 15.78	51.56 ± 3.67	46.56 ± 15.73	53.94 ± 3.99	51.34 ± 13.80
InfoGraph	58.22 ± 6.52	53.98 ± 15.87	58.92 ± 6.10	55.66 ± 16.03	59.82 ± 4.87	60.31 ± 9.37	58.09 ± 3.56	56.19 ± 8.57
JOAO	59.57 ± 5.86	57.77 ± 13.79	56.65 ± 6.25	51.75 ± 16.12	54.17 ± 3.83	50.95 ± 11.67	52.56 ± 3.54	49.64 ± 11.60
JOAO v2	57.97 ± 6.19	51.76 ± 16.42	59.92 ± 6.38	56.17 ± 14.69	54.20 ± 4.28	51.21 ± 13.84	52.36 ± 3.32	48.32 ± 13.91
SimGRACE	58.73 ± 6.03	57.43 ± 12.50	59.60 ± 6.12	55.97 ± 13.14	60.70 ± 4.64	60.21 ± 6.83	58.70 ± 4.68	57.93 ± 11.70
GALOPA	84.16 ± 4.48	83.06 ± 5.35	71.70 ± 5.49	70.74 ± 6.18	<u>91.60 ± 2.16</u>	<u>91.82 ± 2.20</u>	<u>89.94 ± 2.24</u>	<u>90.07 ± 2.27</u>
PaGCL	72.07 ± 5.49	69.75 ± 6.57	71.34 ± 5.39	70.38 ± 6.84	85.22 ± 5.72	84.02 ± 8.54	81.11 ± 5.40	80.09 ± 9.06
SMILES	<u>95.95 ± 2.38</u>	<u>95.99 ± 2.45</u>	<u>96.60 ± 1.82</u>	<u>96.48 ± 2.01</u>	90.25 ± 1.79	90.25 ± 1.85	88.74 ± 2.17	88.53 ± 2.41
HGBN	98.57 ± 1.33	98.57 ± 1.34	98.09 ± 1.39	98.02 ± 1.48	98.75 ± 0.80	98.76 ± 0.80	98.78 ± 0.75	98.79 ± 0.76

The similarity computation costs $O(KD_g)$, and the contrastive loss function requires $O(mD_b(n_i + M))$, where M denotes the number of negative samples and n_i is the number of positive samples for the i -th anchor. Thus, the overall time complexity can be expressed as $O(L(N_g D_g^2 + E_g D_g) + L_b(ND_b^2 + ED_b) + KD_g + mD_b(n_i + M))$.

4 Experiments

Datasets. We evaluate HGBN on eight benchmark multi-graph datasets across multiple domains: two molecular datasets (NCI(1), NCI(109)), two text datasets (AICV, AIDB), three protein-related datasets (PROTEINS, ENZYMES, DD), and one molecular property prediction dataset (PCBA). More detailed description and statistical information for these datasets is provided in Appendix B.

Baselines. We compare HGBN with seven supervised and eight self-supervised baselines. The supervised baselines follow the multi-instanced MGL paradigm [35, 36, 38], including instance-space (mi-SVM [1], MI-SVM), bag-space (MInD [8], MILES [7]), and embedding-space methods (Attention [14], G-Attention, B-Graph [21]). For self-supervised baselines, to our knowledge, this is the first study of SSL in the MGL setting. We therefore compare against seven graph SSL methods (GraphCL [44], InfoGraph [24], JOAO [43], JOAO-v2 [43], SimGRACE [40], GALOPA [30], PaGCL [45]) and one SSL MIL method (SMILES [31]). Since these methods are originally designed for single graphs, we adapt them to MGL by first learning graph-level representations and then aggregating them via max-pooling to obtain bag-level representations.

Experimental Setup. All experiments are conducted on a workstation with an Intel(R) Xeon(R) CPU, 250 GB RAM, and an NVIDIA RTX A6000 GPU, running Ubuntu 20.04.2, Python 3.9, and PyTorch 2.0.1. We use the official implementations and default hyperparameters for baselines. For HGBN, we adopt a two-layer graph encoder (hidden size 256) and a two-layer network encoder (hidden size 128). Models are trained with SGD (learning rate 0.01, weight decay

1×10^{-6}). Batch size is chosen from $\{16, 32, 64, 128\}$ depending on dataset size. The temperature τ and threshold σ are fixed at 0.2. Bag-level representations generated by HGBN and all SSL baselines are evaluated with a linear SVM for downstream classification. Additionally, the linear SVM selects the parameter C from $\{0.001, 0.01, 0.1, 1, 10, 100, 1000\}$ via cross-validation on the training set.

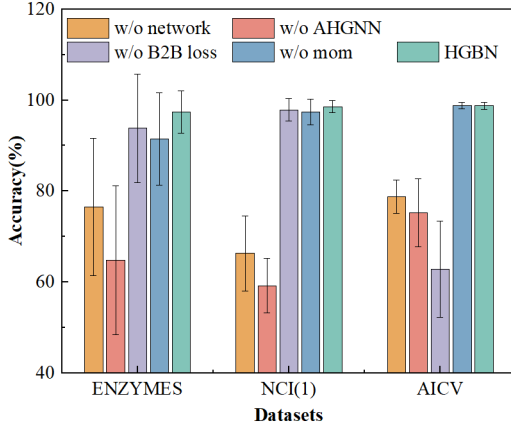
Detailed statistical analyses are provided in Appendix A.

4.1 Comparison Analysis

In this experiment, we compare the proposed HGBN with 15 baseline methods across eight multi-graph datasets using two evaluation metrics, as reported in Tables 1 and 2. HGBN consistently surpasses all supervised baselines, with an average improvement of 24.54% in both accuracy and F1 score, despite its reliance on abundant bag-level labels. This is primarily because existing supervised methods typically adopt graph-to-instance degeneration strategies, reducing MGL to a standard MIL problem. This approach discards intrinsic graph structures and fails to capture both intra-bag and inter-bag dependencies, thereby limiting representational capacity. HGBN also achieves substantial gains over self-supervised graph learning baselines, with an average improvement of 14.63% in both accuracy and F1 score. These methods are designed for single-graph tasks, focusing on extracting discriminative features from individual graphs. While effective for graph-level tasks, they are insufficient for MGL, which requires expressive bag-level representations that integrate hierarchical dependencies across multiple graphs. HGBN overcomes this limitation by explicitly modeling graph-bag interactions within a unified hierarchical framework. HGBN surpasses the state-of-the-art self-supervised MIL baseline, with an average improvement of 5.18% in both accuracy and F1 score. While that method considers relations among instances, it still relies on graph-to-instance degeneration, which discards fine-grained structural information. In contrast, HGBN directly models graph-level

Table 2: Comparative results on PROTEINS, DD, ENZYMES, and PCBA datasets. The best and second-best values in each Column are highlighted in bold and underlined, respectively. ACC denotes accuracy, F1 denotes F1 score, and “-” indicates that the result could not be obtained due to limited computational resources.

Algorithms	PROTEINS		DD		ENZYMES		PCBA	
	ACC (%)	F1 (%)	ACC (%)	F1 (%)	ACC (%)	F1 (%)	ACC (%)	F1 (%)
mi-SVM	70.27 ± 1.83	71.97 ± 2.41	69.77 ± 1.16	65.85 ± 1.85	50.07 ± 0.09	66.70 ± 0.04	63.99 ± 0.49	58.28 ± 0.24
MI-SVM	64.73 ± 2.68	58.21 ± 5.93	66.21 ± 2.68	54.13 ± 7.17	69.33 ± 4.64	48.36 ± 18.47	64.83 ± 0.35	58.72 ± 0.30
MInD	71.42 ± 1.38	69.84 ± 1.51	74.38 ± 1.04	73.55 ± 1.48	69.80 ± 1.99	69.58 ± 1.09	52.13 ± 1.42	53.24 ± 2.09
MILES	67.58 ± 0.28	67.35 ± 1.01	62.42 ± 1.36	57.82 ± 3.22	70.67 ± 3.68	73.10 ± 3.10	-	-
Attention	71.20 ± 1.50	70.70 ± 2.50	63.90 ± 1.50	62.50 ± 2.30	67.20 ± 6.80	70.70 ± 4.00	62.77 ± 0.14	63.24 ± 0.94
G-Attention	72.30 ± 0.90	71.50 ± 2.50	65.80 ± 2.50	65.00 ± 2.90	66.20 ± 6.20	69.30 ± 5.10	64.05 ± 0.85	66.50 ± 2.10
B-Graph	59.64 ± 6.80	75.00 ± 7.07	64.92 ± 11.19	58.52 ± 11.69	50.50 ± 16.70	71.14 ± 13.37	64.00 ± 3.33	72.50 ± 6.50
GraphCL	61.20 ± 10.16	60.93 ± 12.41	60.04 ± 10.09	59.30 ± 12.52	69.10 ± 14.29	65.47 ± 19.34	68.17 ± 1.48	69.01 ± 2.84
InfoGraph	59.97 ± 8.28	59.11 ± 10.76	61.12 ± 9.79	60.12 ± 12.39	68.10 ± 12.37	65.60 ± 17.26	67.32 ± 1.56	68.23 ± 2.13
JOAO	57.43 ± 8.34	57.92 ± 9.77	60.29 ± 9.43	59.95 ± 11.12	64.70 ± 13.06	63.61 ± 17.19	67.07 ± 1.50	68.29 ± 3.55
JOAO v2	59.27 ± 8.15	57.56 ± 11.63	62.12 ± 9.10	61.96 ± 11.74	66.10 ± 16.81	66.44 ± 17.34	68.46 ± 1.88	69.44 ± 2.43
SimGRACE	57.33 ± 9.11	57.13 ± 10.95	64.17 ± 8.66	61.84 ± 11.22	74.20 ± 12.41	72.60 ± 15.26	65.50 ± 2.97	66.45 ± 8.28
GALOPA	85.50 ± 6.87	83.99 ± 7.93	90.08 ± 6.09	90.05 ± 5.69	85.60 ± 10.67	84.84 ± 11.95	-	-
PaGCL	78.27 ± 7.83	79.17 ± 8.63	-	-	76.90 ± 14.61	76.85 ± 17.36	71.02 ± 1.66	71.47 ± 2.47
SMILES	<u>94.33 ± 4.09</u>	<u>94.26 ± 4.20</u>	<u>94.04 ± 4.67</u>	<u>94.10 ± 5.17</u>	<u>88.20 ± 8.57</u>	<u>87.38 ± 11.13</u>	<u>96.00 ± 4.09</u>	<u>95.78 ± 5.95</u>
HGBN	97.10 ± 4.38	96.94 ± 6.15	97.54 ± 3.19	97.54 ± 3.30	97.40 ± 4.63	96.92 ± 6.00	98.95 ± 1.03	98.96 ± 0.98

**Figure 3: The ablation analysis result on the ENZYMES, NCI(1), and AICV multi-graph datasets.**

structures and simultaneously captures hierarchical dependencies across graphs and bags, yielding more expressive and informative bag representations. Overall, these results demonstrate that HGBN effectively preserves fine-grained graph structures, captures hierarchical dependencies, and learns discriminative, generalizable bag-level representations. It also shows an average improvement of 4.82% in both accuracy and F1 score over state-of-the-art baselines, thereby validating the effectiveness of HGBN.

4.2 Ablation Analysis

To evaluate the effectiveness of each proposed component within HGBN, we conduct ablation studies on three multi-graph datasets

of varying sizes and domains: ENZYMES, NCI(1), and AICV. We use accuracy as the evaluation metric, with results shown in Figure 3. We first examine the contribution of the hierarchical representation and AHGNN. In the **w/o network** setting, we remove the network space, effectively treating all graphs as i.i.d. without modeling dependencies among them. In the **w/o AHGNN** setting, we remove both the network and momentum encoders, using only the graph encoder with direct aggregation to obtain bag representations. Both variants result in a substantial decrease in accuracy, confirming that our hierarchical representation and corresponding AHGNN are critical for capturing complex dependencies and enhancing self-supervised MGL performance. Next, we assess the contribution of our cross-hierarchical contrastive loss. In the **w/o B2B loss** setting, we replace our cross-hierarchical objective with a bag-to-bag contrastive loss, comparing only different bag representations. This change reduces performance, indicating that a direct bag-to-graph contrast is essential for preserving hierarchical information and improving embedding quality. Finally, we investigate the necessity of our asymmetric design. In the **w/o mom** setting, we replace the asymmetric structure with a symmetric one where positive and negative pairs are constructed without momentum updates. This symmetric setting consistently lowers classification accuracy, demonstrating that our asymmetric design is crucial for stabilizing training and enhancing the discriminability of the learned representations. Overall, the ablation results validate the necessity of each HGBN component and highlight their joint contribution to achieving superior self-supervised performance in MGL.

4.3 Parameter Analysis

To evaluate the robustness of HGBN, we analyze three key hyperparameters: the training data ratio (δ) for the downstream task, the threshold (σ), and temperature (τ), using accuracy as the evaluation

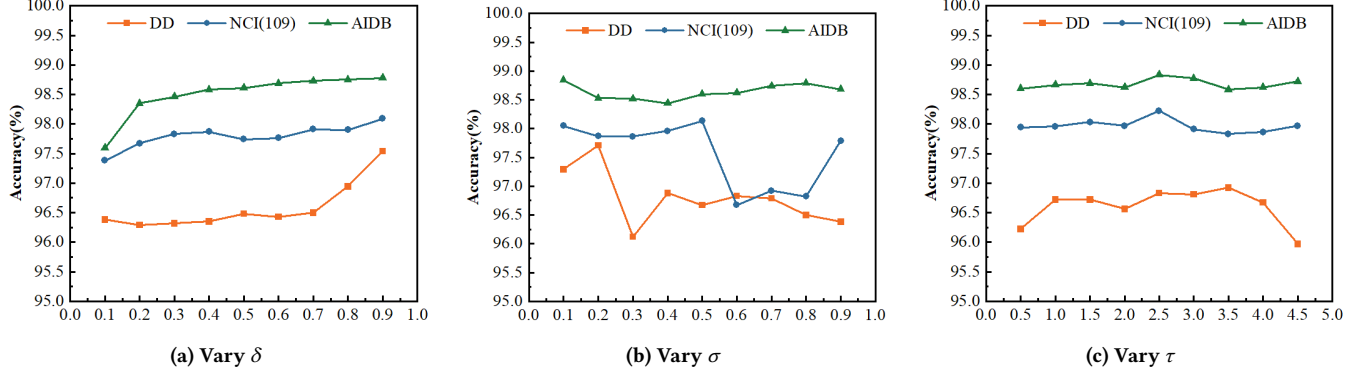


Figure 4: Multi-graph classification performance of HGBN under varying parameters: (a) training data ratio δ , (b) threshold σ , and (c) temperature parameter τ on DD, NCI(109), and AIDB datasets.

Table 3: Runtime comparison results on NCI(109), AICV, PROTEINS, and ENZYMES datasets (in Seconds).

Algorithms	NCI(109)	AICV	PROTEINS	ENZYMES
GraphCL	3.3686	137.02	2.0323	0.7663
InfoGraph	15.069	109.79	7.5454	1.6111
JOAO	6.8350	222.80	2.5178	0.9144
JOAO v2	6.7747	220.24	2.5320	1.0122
SimGRACE	<u>2.8143</u>	<u>79.577</u>	<u>1.4407</u>	0.8017
GALOPA	202.59	361.74	128.11	15.723
PaGCL	249.41	104.19	2543.8	22.672
HGBN	2.6567	72.577	1.3312	0.7335

metric on the DD, NCI(109), and AIDB datasets, as shown in Figure 4. **Train ratio (δ)**. Classification performance generally improves with more training data, as additional samples aid model optimization. However, gains are modest since the SVM classifier largely depends on feature quality. Once robust bag representations are obtained, increasing the train ratio further offers limited improvement, confirming that HGBN produces discriminative and stable representations even for a simple downstream classifier. **Threshold (σ)**. The effect of σ varies across datasets. On AIDB, performance is largely insensitive to σ , reflecting weaker inter-graph dependencies. In contrast, DD and NCI(109) have stronger dependencies, with optimal performance at $\sigma \approx 0.2$ for DD and $\sigma \approx 0.5$ for NCI(109); performance drops outside these ranges. This highlights the importance of σ in constructing informative networks for datasets with significant structural dependencies. **Temperature (τ)**. HGBN shows stable performance across the tested range of τ , with optimal values at $\tau = 3.5$ for DD and $\tau = 2.5$ for NCI(109) and AIDB. Overall accuracy remains largely unaffected, indicating that HGBN is insensitive to τ and confirming its stability. This analysis shows that HGBN delivers consistent and robust performance across diverse hyperparameters, highlighting its broad applicability.

4.4 Efficiency Analysis

To evaluate the computational efficiency of HGBN, we compare its running time with that of seven representative self-supervised

graph learning algorithms: GraphCL, InfoGraph, JOAO, JOAO v2, SimGRACE, GALOPA, and PaGCL. We focus on these methods because their computational complexity is directly comparable to HGBN. In contrast, supervised and self-supervised MIL methods involve additional graph-to-instance processing that would confound a fair comparison. To ensure a fair evaluation, we fix the batch size at 32 across all datasets and record the training and testing time per epoch, excluding data preprocessing and model loading. As shown in Table 3, HGBN demonstrates superior computational efficiency across all datasets. Furthermore, as the dataset size increases, HGBN’s efficiency advantage becomes more pronounced, particularly on larger datasets like AICV. This trend highlights the exceptional scalability of our proposed HGBN framework, making it highly suitable for large-scale applications.

5 Conclusion

In this paper, we tackle the critical problem of self-supervised MGL, which is essential for applications with limited labeled data, such as bioinformatics, chemistry, and computing power networks. We propose the HGBN, a novel self-supervised MGL framework that explicitly models hierarchical dependencies among graphs and bags through a unified graph-bag-network representation. HGBN employs an AHGNN to learn discriminative graph-level and bag-level representations, introduces cross-hierarchical contrastive objectives to capture dependencies across levels, and leverages asymmetric network outputs to generate positive and negative pairs while preserving structural and semantic integrity. Extensive experiments on eight benchmark multi-graph datasets demonstrate that HGBN consistently and significantly outperforms state-of-the-art supervised and self-supervised baselines, validating its effectiveness across diverse domains, including text, bioinformatics, and chemistry. Future work will focus on extending HGBN to enable self-supervised graph representation learning while fine-tuning downstream tasks with limited bag-level supervision.

Acknowledgments

This research is supported by the National Natural Science Foundation of China under Grant No. 62432003.

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A Statistical Analysis

To further assess the effectiveness of HGBN, we perform a comprehensive statistical analysis using the Friedman test [10], followed by the Nemenyi post-hoc test, to examine the significance of performance differences between HGBN and fifteen baseline algorithms across multiple datasets. The Friedman test is applied to both accuracy and F1 scores of all algorithms on eight multi-graph datasets, yielding p -values of 1.3793×10^{-8} and 4.8097×10^{-11} , respectively. Both values are well below the conventional significance threshold of 0.05, leading to the rejection of the null hypothesis that assumes equivalent performance across algorithms. This result confirms that statistically significant differences exist among the algorithms in terms of both accuracy and F1 score.

Subsequently, we employ the Nemenyi test to evaluate pairwise differences among the algorithms. According to the Nemenyi criterion, if the difference in average ranks between two algorithms exceeds the Critical Difference (CD), their performance is considered statistically significant. In Table 2, some baseline algorithms did not produce results on certain large-scale datasets due to computational or time constraints. For instance, PCBA was not evaluated on MILES and GALOPA, and DD was not evaluated on PaGCL. To maintain a conservative comparison, missing results are assigned the best observed performance of 100% under the corresponding metric. With $k = 16$ algorithms and $n = 8$ datasets at a significance level of $\alpha = 0.05$, the critical difference is computed as

$CD = q_\alpha \sqrt{\frac{k(k+1)}{6n}} = 8.1556$. Figs. 5 and 6 illustrate the results of the Nemenyi test for accuracy and F1, respectively. HGBN exhibits statistically significant improvements over the majority of baseline algorithms across both metrics. Moreover, even in cases where the rank differences fall below the critical threshold, HGBN consistently achieves superior performance, further demonstrating its effectiveness in MGL tasks.

B Datasets

In this section, we provide a more detailed description of the datasets, with key characteristics and statistics as shown in Table 4.

NCI(1) and NCI(109). These are two subsets of the NCI cancer screening datasets for anticancer activity prediction. Each chemical compound is represented as a graph where nodes are atoms and edges are bonds. A compound is labeled positive if it exhibits activity against the target cancer. Multi-graph bags are constructed by sampling a combination of positive and negative graphs to form positive bags, and exclusively negative graphs for negative bags.

AICV and AIDB. These are two text datasets derived from the DBLP dataset [25]. Papers published in three fields: artificial intelligence (AI), computer vision (CV), and databases (DB) are used to form the datasets. We use a fuzzy cognitive map (E-FCM) [22] approach to extract keywords from paper abstracts and the correlations between them to build a graph. AI is defined as the positive class, while CV and DB are the negative classes, forming the AICV and AIDB datasets, respectively.

PROTEINS and ENZYMES. Both are protein classification datasets [4]. PROTEINS classifies proteins as enzymes and non-enzymes, with nodes representing secondary structures. ENZYMES

classify proteins based on their enzyme class, with each graph representing a single protein. Multi-graph bags are constructed by sampling from both positive and negative classes for positive bags and exclusively from negative classes for negative bags.

DD. The DD dataset [9] is a protein classification dataset comprising enzymes and non-enzymes. Each node represents an amino acid, with an edge between spatially close nodes. Positive bags are formed by sampling a subset of enzymes together with non-enzymes, while negative bags are formed from subsets of non-enzymes.

PCBA. The PCBA dataset is derived from the ogbg-molpcba dataset in the OGB database, originating from MoleculeNet and among the largest MoleculeNet datasets. Each graph represents a molecule, with nodes as atoms and edges as chemical bonds. Each node includes a feature vector. Multi-graph bags are constructed by randomly selecting a mix of positive and negative graphs for positive bags, and only negative graphs for negative bags.

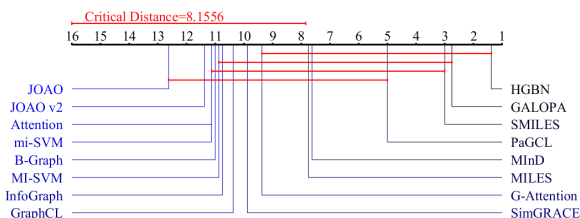


Figure 5: Critical difference diagram of HGBN and fifteen competing algorithms based on accuracy (Nemenyi test).

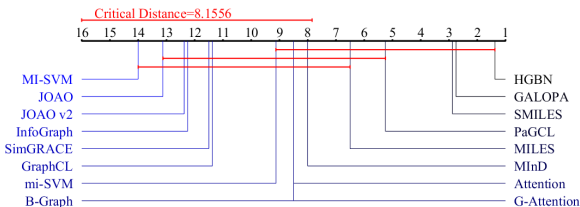


Figure 6: Critical difference diagram of HGBN and fifteen competing algorithms based on F1 score (Nemenyi test).

Table 4: Multi-graph datasets.

Datasets	NCI(1)	NCI(109)	AICV	AIDB
#Bags	740	800	2000	2000
#Graphs	4244	3815	12767	12937
#Nodes	117756	105610	254514	257688
#Edges	254326	228116	2178648	2321219
Datasets	PROTEINS	DD	ENZYMES	PCBA
#Bags	300	240	100	10000
#Graphs	1315	1144	297	45050
#Nodes	53819	349810	9880	1156566
#Edges	202242	883282	18631	2500476